

Hierarchical Crossed Cube: A New Interconnection Topology *

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Abstract

An interconnection network plays a critical role of a multi-computer because the system performance is deeply dependent on network latency and throughput. There are a lot of mutually conflicting requirements in designing the topology of interconnection networks. It is almost impossible to design a network which is optimum in all perspectives. Therefore, designing new interconnection networks is still an attractive research. In this paper, we introduce a new interconnection topology, Hierarchical Crossed Cubes(HCC). This topology is suitable for the design of massively parallel systems with thousands of processors. An interesting property of this network is the low vertex degree, which enhances the VLSI design and fabrication of the system. Other pleasing features include symmetry and logarithmic diameter, which imply easy and fast algorithms for communication. Moreover, the HCC is scalable to embed HCC's of lower dimensions. The paper introduces a simple routing algorithm for HCC and proves the connectivity equal to the degree of HCC which implies the maximum fault tolerance.

Keywords: Interconnection topology, Hypercube, Crossed Cube, Hierarchical Crossed Cube, connectivity, routing algorithm, diameter.

1 Introduction

For the graph definition and notation we follow [4]. $G = (V, E)$ is a graph if V is a finite set and E is a subset of $\{(u, v) \mid (u, v) \text{ is an unordered pair of } V\}$. We say that V is the *vertex set* and E is the *edge set*. Two vertices u and v are *adjacent* if $(u, v) \in E$. A path is a sequence of adjacent vertices, written as $\langle v_0, v_1, v_2, \dots, v_m \rangle$, in which all the vertices $v_0, v_1, \dots, v_m$ are distinct except possibly $v_0 = v_m$. For this path, v_0 and v_m are called *end vertices*, and $v_i, 1 \leq i \leq m-1$, is called *internal vertex*. For convenience, we use (u, v) -path to denote the path with end vertices u and v . We also write the path $\langle v_0, P, v_m \rangle$ or $P(v_0, v_m)$, where $P = \langle v_0, v_1, \dots, v_m \rangle$. The *length* of a path P , $len(P)$, is the number of edges in P . n (u, v) -paths $P_1, P_2, \dots, P_n$ are said to be *internally disjoint* if they have no common internal vertices. A vertex cut of G is a set of vertices $S \subseteq G$ such that $G - S$ is disconnected. The connectivity of G , $\kappa(G)$, is the minimum cardinality over all vertex cuts of G . If $\kappa(G) = n$ then G is n -connected. By Menger's Theorem, if $\kappa(G) = n$ then there exist n internally disjoint (u, v) -paths over all pair of vertices $u, v \in V(G)$; and for any set $\{w_1, w_2, \dots, w_n\}$ of vertices in $G - \{u\}$ there are n disjoint (except u) (u, w_i) -paths in G . The *distance* between two distinct vertices u and v of G , denoted by $d_G(u, v)$, is the length of the shortest (u, v) -path of G . The diameter of G , denoted by $Diam(G)$, is defined as $Diam(G) = \max\{d_G(u, v) : u, v \in V(G)\}$.

It is important to design parallel computers using an interconnection network topology that can scale up to a large number of processors and that is capable of sustaining fast communication and data sharing among processors. There are a lot of mutually conflicting requirements in designing the

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topology of interconnection networks. It is almost impossible to design a network which is optimum in all perspectives. Therefore, designing new interconnection networks is still an attractive research. This investigation will continue for decades since parallel and distributed computers are the main solution for the computational problems that will challenge human beings in the twenty-first century.

Hierarchical interconnection networks [12, 13, 16, 11, 17] have attracted considerable attention in the research for multiprocessor systems. A hierarchical design approach allows the network to be constructed incrementally, starting from one or more basic modules. Hierarchical interconnection networks are intuitively appealing when massively parallel processors are to be connected. The Hypercube [14, 2] is one of the most widely used topologies because it has charming properties such as strong connectivity, recursive interconnection, simple routing, a good link complexity, and embedding of various interconnection networks. A variety of Hypercube based hierarchical interconnection networks such as Crossed Cube [6, 7, 8], Enhanced Cube [15], Extended Cube [12], Folded Cube [1], Generalized Hypercube [3], Möbius Cube [5], Twisted Cube [9, 10], and so on have been proposed.

Motivated by the architecture of hierarchical Interconnection network and many noticeable features of Crossed cube, we propose a new interconnection network – the Hierarchical Crossed cube (*HCC*) in this paper. Theoretical network properties such as the diameter, routing algorithm, connectivity will be discussed in detail.

The paper is organized as follows: next section introduces the necessary preliminaries and the formal definition of Hierarchical Crossed Cube. In section 3, we study the routing algorithm and the diameter of *HCC*. The connectivity of *HCC* is proved in Section 4. Finally, we give a conclusion.

2 Preliminary

In this section, we will introduce the formal definition of the Hierarchical Crossed Cubes. To define the Hierarchical Crossed cubes, we need the definitions of Hypercubes [14] and Crossed cubes [6]. For convenience, k and n are both positive integers and $k, n \geq 1$ in the following discussion.

Definition 1 *The n -dimensional Hypercube, Q_n , is a graph $G = (V, E)$ with the vertex set $V =$*

$\{b_{n-1}b_{n-2} \dots b_1b_0 \mid b_i \in \{0, 1\} \text{ for all } 0 \leq i \leq n-1\}$ and the edge set $E = \{(u, v) \mid u \text{ and } v \text{ differ exactly one bit}\}$.

To define Crossed Cubes, as the proposed by Efe [6], the notion so called “pair related” relation is introduced.

Definition 2 *Let R = $\{(00, 00), (10, 10), (01, 11), (11, 01)\}$. Two dibit binary strings $u = u_1u_0$ and $v = v_1v_0$ are pair related, denoted as $u \sim v$, if and only if $(u, v) \in R$.*

The following is the recursive definition of the n -dimensional Crossed cube CQ_n .

Definition 3 [6] *The Crossed cube CQ_1 is a complete graph with two vertices labelled by 0 and 1, respectively. For $n \geq 2$, an n -dimensional Crossed cube CQ_n consists of two $(n-1)$ -dimensional sub-Crossed cubes, CQ_{n-1}^0 and CQ_{n-1}^1 , and a perfect matching between the vertices of CQ_{n-1}^0 and CQ_{n-1}^1 according to the following rule:*

Let $V(CQ_{n-1}^0) = \{0u_{n-2}u_{n-3} \dots u_0 : u_i = 0 \text{ or } 1\}$ and $V(CQ_{n-1}^1) = \{1v_{n-2}v_{n-3} \dots v_0 : v_i = 0 \text{ or } 1\}$. The vertex $u = 0u_{n-2}u_{n-3} \dots u_0 \in V(CQ_{n-1}^0)$ and the vertex $v = 1v_{n-2}v_{n-3} \dots v_0 \in V(CQ_{n-1}^1)$ are adjacent in CQ_n if and only if

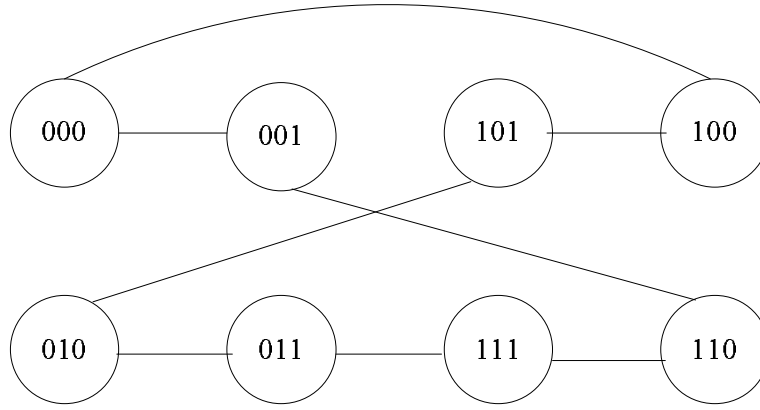
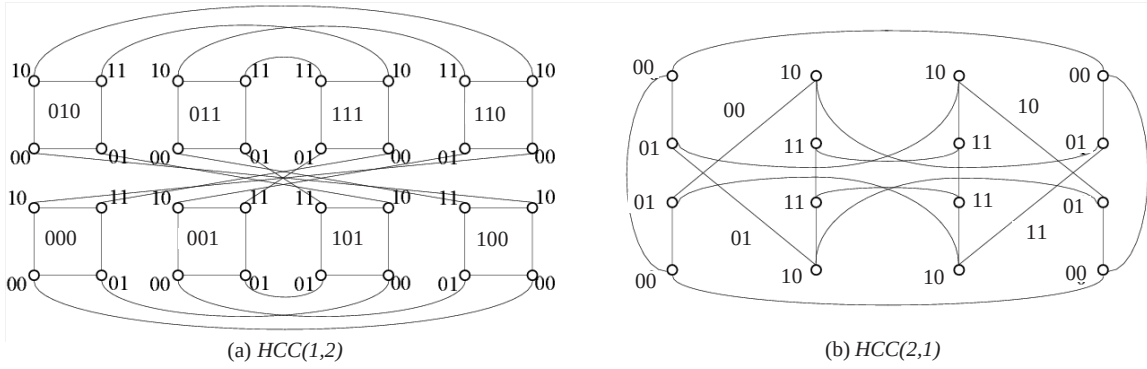
- (1) $u_{n-2} = v_{n-2}$ if n is even, and
- (2) $(u_{2i+1}u_{2i}, v_{2i+1}v_{2i}) \in R$, for $0 \leq i < \lfloor \frac{n-1}{2} \rfloor$.

We are now ready to define the Hierarchical Crossed Cubes.

Definition 4 *Given two positive integers $k, n \geq 1$. A Hierarchical Crossed Cube, $HCC(k, n)$, is a graph $G = (V, E)$ on 2^{k+2n} vertices, where $V = \{b_{k+2n-1}b_{k+2n-2} \dots b_1b_0 \mid b_i \in \{0, 1\}, 0 \leq i \leq k+2n-1\}$ and $E = E_{int} \cup E_{ext}$. The label of a vertex u is divided into three parts, say $u = u_X u_Y u_Z$, where $u_X = u_{k+2n-1}u_{k+2n-2} \dots u_{2n}$, $u_Y = u_{2n-1}u_{2n-2} \dots u_n$, and $u_Z = u_{n-1}u_{n-2} \dots u_0$. The set of edges E is the union of two sets E_{int} and E_{ext} , which are the sets of internal and external edges, respectively, as the following equations.*

- (1) $E_{int} = \{(u, v) \mid u_X = v_X, u_Y = v_Y, \text{ and } (u_Z, v_Z) \in E(CQ_n)\}$ and
- (2) $E_{ext} = \{(u, v) \mid u_Y = v_Z, u_Z = v_Y, \text{ and } (u_X, v_X) \in E(Q_k)\}$.

Figures 1 and 2 illustrate the $HCC(1, 1)$, $HCC(2, 1)$, and $HCC(2, 1)$, respectively.


 Figure 1: $HCC(1, 1)$

 Figure 2: Illustration of $HCC(1, 2)$ and $HCC(2, 1)$

By the definition of Hierarchical Crossed Cubes, the structure of a $HCC(k, n)$ consists of three levels of hierarchy. At the lowest level of hierarchy, we have a pool of 2^{k+2n} vertices. These vertices are grouped into clusters of 2^n vertices each, and the vertices in each cluster are connected to form an CQ_n . For convenience, let $dec(b)$ be the decimal number of the binary string b , let $Ccube$ denote the cluster of CQ_n , and let $Ccube(i, j)$ denote the induced subgraph of $HCC(k, n)$ whose vertex set is $\{u = u_X u_Y u_Z : dec(u_X) = i, dec(u_Y) = j, \text{ and } u_Z = u_{n-1} u_{n-2} \dots u_1 u_0, u_m \in \{0, 1\}, 0 \leq m \leq n-1\}$. Then, 2^n disjoint Ccubes constitute the second level hierarchy, called $CGroup$. Again, let $CGroup(i)$ denote the induced subgraph of $HCC(k, n)$ whose vertex set is $\{u : u \in V(Ccube(i, j)), 0 \leq j \leq 2^{n-1}\}$. Finally, 2^k CGroups are connected in a Hypercube fashion to form a $HCC(k, n)$. Clearly, edges of the Ccubes are called internal edges, and edges be-

tween CGroups are referred to as external edges. The Ccubes in each CGroup have no link to connect each other. The following properties are some observations on $HCC(k, n)$.

Property 1 $HCC(k, n)$ is $(n+k)$ -regular and has 2^{2n+k} vertices for $k, n \geq 1$.

Given an neighbor v of a vertex u in $HCC(k, n)$ with that i is the left most different bit between u and v , then v is called the i -neighbor of u , denoted by $v = N_i(u)$. If $2n \leq i \leq 2n + k - 1$, then $N_i(u)$ is an *external neighbor* of u in $HCC(k, n)$, denoted by $N_{i-2n}^{ext}(u)$. If $0 \leq i \leq n - 1$, then $N_i(u)$ is an *internal neighbor* of u in $HCC(k, n)$, denoted by $N_i^{int}(u)$. Clearly, there are k external neighbors and n internal neighbors for each vertex in $HCC(k, n)$. Let $N^{ext}(u) = \{N_i^{ext}(u) : 0 \leq i < k\}$ and $N^{int}(u) = \{N_i^{int}(u) : 0 \leq i < n\}$.

Lemma 1 Let u be any vertex in $HCC(k, n)$ with $k, n \geq 1$ and $a, b \in N^{ext}(u)$ be two distinct vertices. Then $a_X \neq b_X$.

Lemma 2 Let u and v be two distinct vertices in the same subgraph $Ccube(i, j)$ of $HCC(k, n)$ with $k, n \geq 1$ and $i, j \geq 0$. For each pair of vertices $a \in N^{ext}(u)$ and $b \in N^{ext}(v)$, $a_Y \neq b_Y$.

Lemma 3 The 2^n vertices of $Ccube(i, j)$ connect to exactly 2^n Ccubes in a $CGroup(i')$ if $(i, i') \in E(Q_k)$.

In other words, if $(i, i') \in E(Q_k)$, there is a one-to-one and onto matching between the vertices in $Ccube(i, j)$ and the Ccubes in $CGroup(i')$. Let $HCC^i(k, n)$, $i \in \{0, 1\}$, be the subgraph of $HCC(k, n)$ induce by the vertices which have the label with the left most bit is i . Clearly, $HCC^i(k, n)$ is isomorphic to $HCC(k - 1, n)$.

Property 2 $HCC(k, n)$ with $k \geq 2$ and $n \geq 1$ can be constructed from two $HCC(k - 1, n)$ s, say $HCC^0(k, n)$ and $HCC^1(k, n)$, by a perfect matching between the vertices of $HCC^0(k, n)$ and $HCC^1(k, n)$.

3 The routing algorithm

Herein, we need the shortest path routing algorithms of Hypercubes[14] and Crossed cubes[7] to build the routing algorithm of $HCC(k, n)$. The shortest path routing algorithm of Q_k [14] and the shortest path routing algorithm of CQ_n , proposed by Efe [7], are illustrated as Algorithms 1 and 2, respectively. Then, the routing algorithm of $HCC(k, n)$ is illustrated as Algorithm 3.

Lemma 4 By the routing algorithm 3, the diameter of $HCC(k, n)$, $Diam(HCC(k, n)) \leq 2\lceil \frac{n+1}{2} \rceil + k$ for $k \geq 2$ and $n \geq 1$, and $Diam(HCC(1, n)) \leq 2\lceil \frac{n+1}{2} \rceil + 2$ for $n \geq 1$.

4 The connectivity

In this section, we will prove the connectivity of $HCC(k, n)$ to be $k + n$. Before studying the connectivity of $HCC(k, n)$, we show that $HCC(1, n)$ is $(n + 1)$ -connected as follows.

Lemma 5 $HCC(1, n)$ is $(n + 1)$ -connected for $n \geq 1$.

Proof. $HCC(1, n)$ consists of $CGroup(0)$ and $CGroup(1)$, and each includes 2^n Ccubes. To prove $HCC(1, n)$ to be $(n + 1)$ -connected, we will show there are $n + 1$ internally disjoint (u, v) -paths

for any two distinct vertices u and v of $HCC(1, n)$. In the following, we divide the proof into three cases according to the positions of u and v .

Case 1: $u_X = v_X$ and $u_Y = v_Y$.

In other words, u and v are both in the same $Ccube(i, j)$. Without loss of generality, let $u, v \in V(Ccube(0, 0))$. Since CQ_n is n -connected, there are n internally disjoint (u, v) -paths in $Ccube(0, 0)$. Let $a = N_0^{ext}(u)$ and $b = N_0^{ext}(v)$. By Lemma 2, $a_Y \neq b_Y$. Hence, a and b are in two different Ccubes of $CGroup(1)$. Suppose a and b are in $Ccube(1, j)$ and $Ccube(1, j')$, respectively. We can choose two vertices c in $Ccube(1, j)$ and d in $Ccube(1, j')$ such that $c_Z = d_Z \neq u_Y$. Let $dec(c_Z) = l$. Thus, $N_0^{ext}(c)$ and $N_0^{ext}(d)$ are both in the same $Ccube$, say $Ccube(0, l)$. Therefore, we can construct the (a, c) -path P_1 in $Ccube(1, j)$, the (b, d) -path P_2 in $Ccube(1, j')$, and the $(N_0^{ext}(c), N_0^{ext}(d))$ -path P_3 in $Ccube(0, l)$. Then, $\langle u, a, P_1, c, N_0^{ext}(c), P_3, N_0^{ext}(d), d, P_2, v \rangle$ is the $(n + 1)$ th internally disjoint (u, v) -path of $HCC(1, n)$. See Figure 3 for illustration.

Case 2: $u_X = v_X$ and $u_Y \neq v_Y$.

In other words, u and v are in different Ccubes but in the same $CGroup(i)$, say $i = 0$. Without loss of generality, let $u \in V(Ccube(0, j))$ and $v \in V(Ccube(0, j'))$. Let $S_1 = \{x_1, x_2, \dots, x_l = u, \dots, x_{n+1}\}$ and $S_2 = \{y_1, y_2, \dots, y_{l'} = v, \dots, y_{n+1}\}$ be two vertex sets in $Ccube(0, j)$ and $Ccube(0, j')$, respectively, such that $(x_i)_Z = (y_i)_Z$ for $1 \leq i \leq n + 1$. Since $Ccube(0, j)$ and $Ccube(0, j')$ are both isomorphic to CQ_n , we can find the two sets S_1 and S_2 by choosing the same vertices in CQ_n . Since CQ_n is n -connected, there are n disjoint(except u) (u, x_i) -paths, P_i , for $1 \leq i \neq l \leq n + 1$ in $Ccube(0, j)$ and there are also n disjoint(except v) (y_i, v) -paths, P'_i , for $1 \leq i \neq l' \leq n$ in $Ccube(0, j')$. Note that $N_0^{ext}(x_i)$ and $N_0^{ext}(y_i)$ are in the same $Ccube$. Let R_i be the $(N_0^{ext}(x_i), N_0^{ext}(y_i))$ -path in $Ccube(1, dec((x_i)_Z))$. Thus, we can construct $n + 1$ internally disjoint (u, v) -paths of $HCC(1, n)$ with the format $\langle u, P_i, x_i, N_0^{ext}(x_i), R_i, N_0^{ext}(y_i), y_i, P'_i, v \rangle$, $1 \leq i \leq n + 1$. Note that $len(P_l) = 0$ and $len(P'_{l'}) = 0$. See Figure 4 for illustration.

Case 3: $u_X \neq v_X$.

That is, u and v are in different $CGroups$. Without loss of generality, let u and v belong to $Ccube(0, j)$ and $Ccube(1, j')$, respectively. Let $S_1 = \{x_1, x_2, \dots, x_l = u, \dots, x_{n+1}\}$ and $S_2 = \{y_1, y_2, \dots, y_{l'} = v, \dots, y_{n+1}\}$ be two vertex sets in $Ccube(0, j)$ and $Ccube(1, j')$, respectively, such that $(x_{n+1})_Z = v_Y$, $(y_{n+1})_Z = u_Y$, and

Algorithm 1 Procedure $Q_k\text{Routing}(u,v)$ // A shortest path routing algorithm of Q_k .

- 1: $l \leftarrow \text{LeftMostDifferingBitIndex}(u,v)$
 - 2: route to l -neighbor
-

Algorithm 2 Procedure $CQ\text{Routing}(u,v)$ // The shortest path routing algorithm of CQ_n . [7]

- 1: $l \leftarrow \text{LeftMostDifferingBitIndex}(u,v)$
 - 2: **if** (l is odd) and $(u_{l-1} \neq v_{l-1})$ **then**
 - 3: $d \leftarrow \text{choice}(\{l, l-1\})$
 - 4: route to d -neighbor
 - 5: **else if** there exist $\alpha \leftarrow u_{2k+1}u_{2k}$, $\beta \leftarrow v_{2k+1}v_{2k}$ such that $(\alpha \not\sim \beta)$ and $l > 2k+1$. **then**
 - 6: $d \leftarrow \text{choice}(\text{MOVE}(\alpha, \beta))$
 - 7: route to d -neighbor
 - 8: **else**
 - 9: route to l -neighbor $\{u$ and v are adjacent.}
 - 10: **end if**
-

Algorithm 3 $HCC\text{Routing}(u,v,k,n)$ // u the source vertex and v the destination vertex

- 1: **if** $d_{Q_k}(u_X, v_X)$ is even **then**
 - 2: route $u_X u_Y u_Z$ to $u_X u_Y v_Z$ by CQ_n routing method on u_Z to v_Z .
 - 3: route $u_X u_Y v_Z$ to $u'_X v_Z u_Y$ by the first step of Hypercube routing method on u_X to v_X , and automatic swap u_Y and v_Z .
 - 4: route $u'_X v_Z u_Y$ to $u'_X v_Z v_Y$ by CQ_n routing method on u_Y to v_Y .
 - 5: route $u'_X v_Z v_Y$ to $v_X v_Y v_Z$ by Hypercube routing method on u'_X to v_X .
 - 6: **else if** $d_{Q_k}(u_X, v_X)$ is odd **then**
 - 7: route $u_X u_Y u_Z$ to $u_X u_Y v_Y$ by CQ_n routing method on u_Z to v_Y .
 - 8: route $u_X u_Y v_Y$ to $u'_X v_Y u_Y$ by the first step of Hypercube routing method on u_X to v_X , and automatic swap u_Y and v_Y .
 - 9: route $u'_X v_Y u_Y$ to $u'_X v_Y v_Z$ by CQ_n routing method on u_Y to v_Z .
 - 10: route $u'_X v_Y v_Z$ to $v_X v_Y v_Z$ by Hypercube routing method on u'_X to v_X .
 - 11: **else if** $d_{Q_k}(u_X, v_X) = 0$ // $u_X = v_X$ **then**
 - 12: **if** $u_Y = v_Y$ **then**
 - 13: route u_Z to v_Z by using the CQ_n routing method.
 - 14: **else**
 - 15: route $u_X u_Y u_Z$ to $u'_X u_Z u_Y$ by change the left most bit of u_X .
 - 16: route $u'_X u_Z u_Y$ to $u'_X u_Z v_Y$ by CQ_n routing method on u_Y to v_Y .
 - 17: route $u'_X u_Z v_Y$ to $u_X v_Y u_Z$ by change the left most bit of u'_X .
 - 18: route $u_X v_Y u_Z$ to $v_X v_Y v_Z$ by CQ_n routing method on u_Z to v_Z .
 - 19: **end if**
 - 20: **end if**
-

Table 1: The MOVE table.[7]

α/β	00	01	10	11
00	ϕ	$\{l\}$	$\{2k+1, l\}$	$\{2k\}$
01	$\{2k\}$	$\{2k+1, l\}$	$\{l\}$	ϕ
10	$\{2k+1, l\}$	$\{2k\}$	ϕ	$\{l\}$
11	$\{l\}$	ϕ	$\{2k\}$	$\{2k+1, l\}$

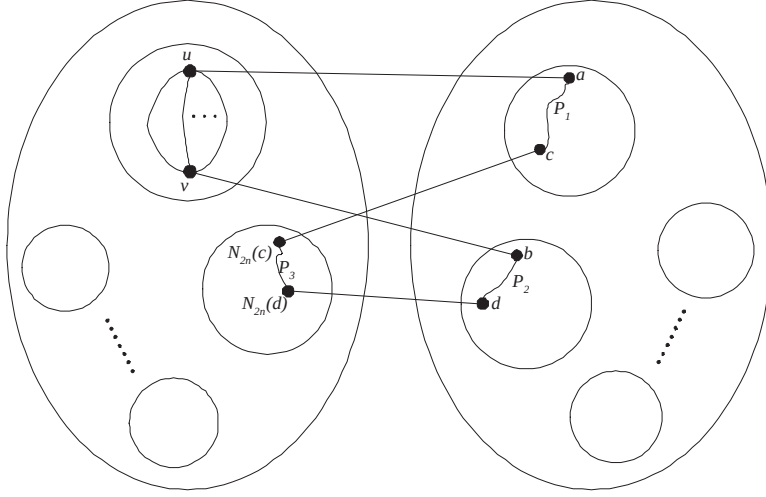


Figure 3: The proof of case 1 of Lemma 5.

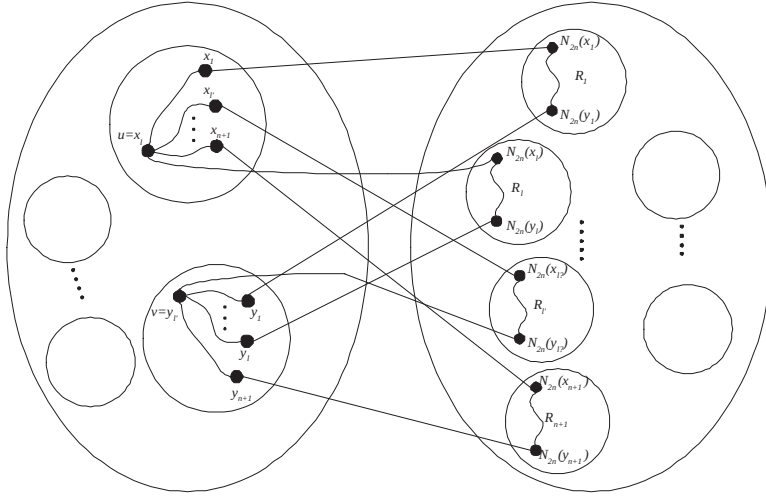


Figure 4: The proof of case 2 of Lemma 5.

$(x_{n+1}, y_{n+1}) \in E(HCC(k, n))$. Note that it is possible that $u = x_{n+1}$ or $v = y_{n+1}$. It implies that the external neighbor of u (v , respectively) maybe is in $Ccube(1, j')$ ($Ccube(0, j)$, respectively). Since CQ_n is n -connected, there are n disjoint(except u) (u, x_i) -paths, P_i , for

$1 \leq i \neq l \leq n+1$ in $Ccube(0, j)$ and there are also n disjoint(except v) (y_i, v) -paths, P'_i , for $1 \leq i \neq l' \leq n$ in $Ccube(1, j')$. Let $x'_i = N_0^{ext}(x_i)$ and $l_i = dec((x'_i)_Y)$. Therefore, we can choose a vertex w_i in $Ccube(1, l_i)$ for $1 \leq i \leq n$ such that $(w_i)_Z = (y_i)_Z$. Let $y'_i = N_0^{ext}(y_i)$ and

$l'_i = \text{dec}((y'_i)_Y)$. Let R_i be the (x'_i, w_i) -path in $Ccube(1, l_i)$ and R'_i be the $(N_0^{ext}(w_i), y'_i)$ -path in $Ccube(0, l'_i)$ for all $1 \leq i \leq n$. Therefore, we can construct $n + 1$ internally disjoint (u, v) -paths of $HCC(1, n)$ with the format $\langle u, P_i, x_i, x'_i, R_i, w_i, N_0^{ext}(w_i), R'_i, y'_i, y_i, P'_i, v \rangle$, $1 \leq i \leq n$, and the format $\langle u, P_{n+1}, x_{n+1}, y_{n+1}, P'_{n+1}, v \rangle$. See Figure 5 for illustration. $\square$

Lemma 6 *Given two positive integers $k \geq 2$ and $n \geq 1$. If $HCC(k-1, n)$ is $(n+k-1)$ -connected, then $HCC(k, n)$ is $(n+k)$ -connected.*

Proof. To prove this Lemma, we need to construct $(n+k)$ internally disjoint (u, v) -paths for any two distinct vertices u and v in $HCC(k, n)$. Then, we divide the proof into two cases based on the positions of u and v .

Case 1: u and v are both in the same subgraph $HCC^i(k, n)$, $i = 0, 1$, of $HCC(k, n)$.

Without loss of generality, let u and v both belong to $HCC^0(k, n)$. Since $HCC(k-1, n)$ is $(n+k-1)$ -connected, there are $(n+k-1)$ internally disjoint (u, v) -paths in $HCC^0(k, n)$. Let P be a path joining $N_{k-1}^{ext}(u)$ and $N_{k-1}^{ext}(v)$ in $HCC^1(k, n)$. Then $\langle u, N_{k-1}^{ext}(u), P, N_{k-1}^{ext}(v), v \rangle$ is the $(n+k)$ th internally disjoint (u, v) -path in $HCC(k, n)$. See Figure 6 for illustration.

Case 2: u and v are in $HCC^i(k, n)$ and $HCC^{1-i}(k, n)$, $i = 0, 1$, respectively.

Without loss of generality, let u and v belong to $HCC^0(k, n)$ and $HCC^1(k, n)$, respectively. Next, we consider the two conditions: $v = N_{k-1}^{ext}(u)$ and $v \neq N_{k-1}^{ext}(u)$.

Subcase 2.1: $v = N_{k-1}^{ext}(u)$.

Since there are 2^{k+2n-1} edges between $HCC^0(k, n)$ and $HCC^1(k, n)$, we can choose $(k+n-1)$ edges, (w_i, w'_i) for $1 \leq i \leq k+n-1$, between $HCC^0(k, n)$ and $HCC^1(k, n)$ except (u, v) . Without loss of generality, let w_i and w'_i belong to $HCC^0(k, n)$ and $HCC^1(k, n)$, respectively. Since $HCC(k-1, n)$ is $(n+k-1)$ -connected, there are $(n+k-1)$ disjoint(except u) (u, w_i) -paths, P_i , in $HCC^0(k, n)$, and there are also $(n+k-1)$ disjoint(except v) (v, w'_i) -paths, P'_i , in $HCC^1(k, n)$ for $1 \leq i \leq n+k-1$. Thus, there exist $(n+k-1)$ internally disjoint (u, v) -paths in $HCC(k, n)$ with the format $\langle u, P_i, w_i, w'_i, P'_i, v \rangle$, $1 \leq i \leq n+k-1$. Finally, $\langle u, v \rangle$ is the $(n+k)$ th internally disjoint (u, v) -path in $HCC(k, n)$. See Figure 7 for illustration.

Subcase 2.2: $v \neq N_{k-1}^{ext}(u)$.

Let $v' = N_{k-1}^{ext}(v)$. Since $HCC(k-1, n)$ is $(n+k-1)$ -connected, there $(n+k-1)$ inter-

nally disjoint (u, v') -paths, P_i , in $HCC^0(k, n)$. Let $P_i = \langle u, P'_i, w_i, v' \rangle$ for $1 \leq i \leq n+k-1$ and $w'_j = N_{k-1}^{ext}(w_j)$ for $1 \leq j \leq n+k-2$. If there are ambiguities, we write $w'_{n+k-1} = N_{k-1}^{ext}(u)$. Since $HCC(k-1, n)$ is $(n+k-1)$ -connected, there are also $n+k-1$ disjoint(except v) (v, w'_i) -paths, R_i , in $HCC^1(k, n)$ for $1 \leq i \leq n+k-1$. As a consequence, we can build $k+n$ internally disjoint (u, v) -paths in $HCC(k, n)$ with the three formats $\langle u, P'_i, w_i, w'_i, R_i, v \rangle$ for $1 \leq i \leq n+k-2$, $\langle u, P_{n+k-1}, v', v \rangle$, and $\langle u, w'_{n+k-1}, R_{n+k-1}, v \rangle$. See Figure 8 for illustration. $\square$

Theorem 1 *Given two positive integers $k, n \geq 1$. The connectivity of (k, n) -Hierarchical Crossed Cube, $\kappa(HCC(k, n)) = n+k$.*

Proof. Note that $HCC(k, n)$ is $(n+k)$ -regular graph. Then $\kappa(HCC(k, n)) \leq n+k$. By Lemmas 5 and 6, we know that $HCC(k, n)$ is $(n+k)$ -connected. That is, $\kappa(HCC(k, n)) \geq n+k$. Hence, $\kappa(HCC(k, n)) = n+k$. $\square$

Since $HCC(k, n)$ is $(n+k)$ -connected and $(n+k)$ -regular, $HCC(k, n)$ has optimal connectivity.

5 Conclusion

In this paper, we introduce a new interconnection topology, Hierarchical Crossed Cubes, $HCC(k, n)$. We also show a simple routing algorithm and the connectivity of $HCC(k, n)$. $HCC(k, n)$ has many interesting properties such as low degree, logarithmic diameter, simple routing algorithm, and maximum connectivity. Other properties, such as fault diameter, hamiltonian and fault-tolerant hamiltonian, panconnectivity etc. will be investigated in the future.

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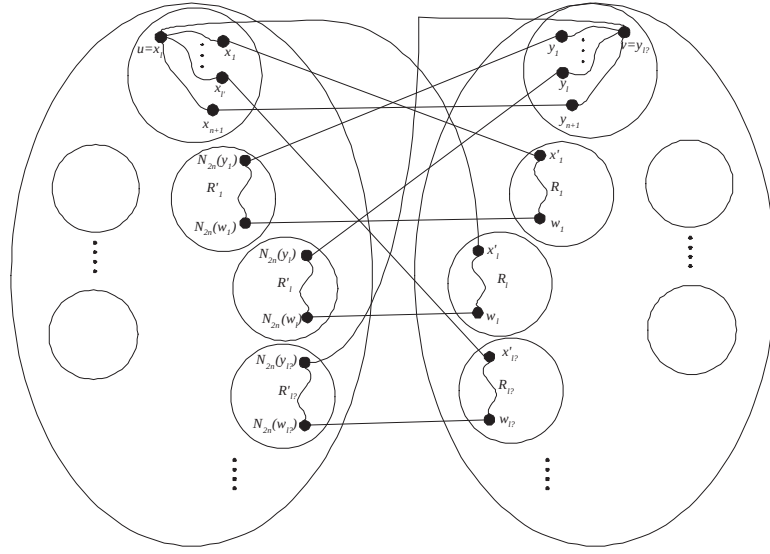


Figure 5: The proof of case 3 of Lemma 5.

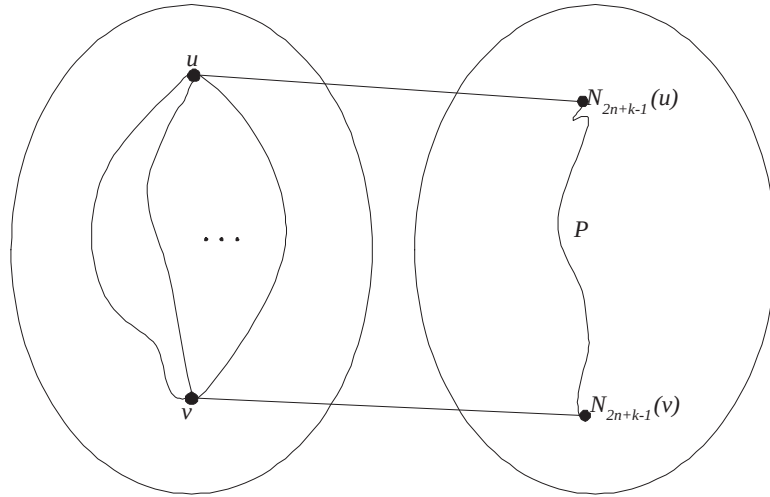


Figure 6: The proof of case 1 of Lemma 6.

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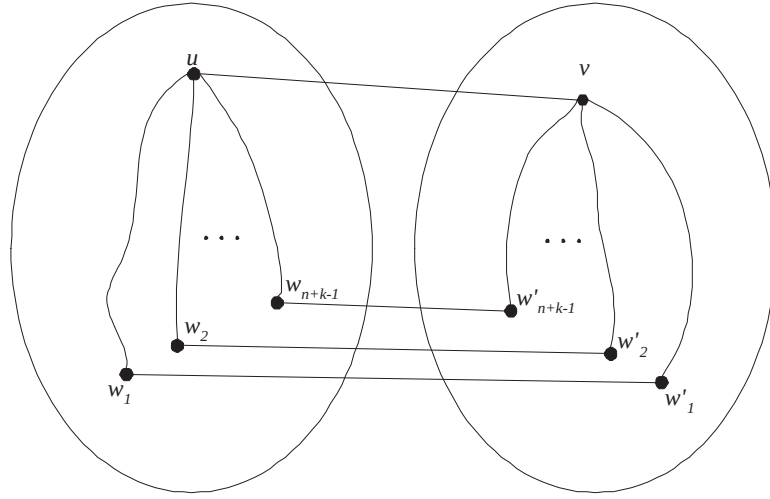


Figure 7: The proof of subcase 2.1 of Lemma 6.

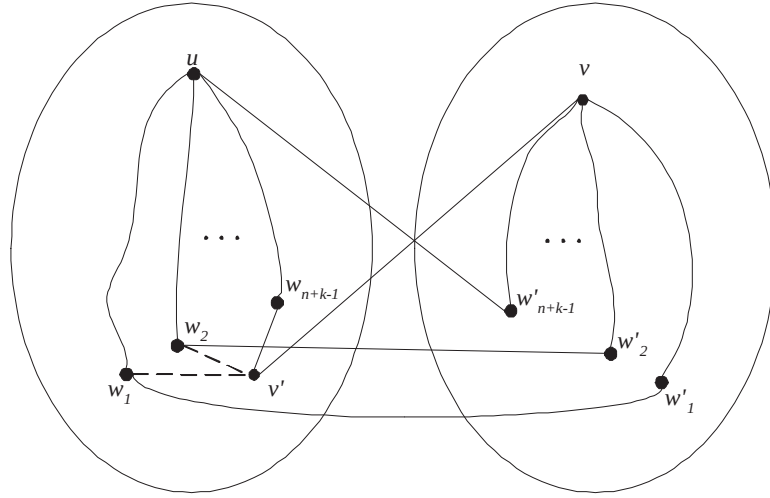


Figure 8: The proof of subcase 2.2 of Lemma 6.

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